The goal of this work is to fine-tune a Vision-Language Model (VLM) for converting handwritten mathematical formulas into LaTeX code. The task was evaluated under different inference settings (zero-shot, one-shot, and supervised fine-tuning) using publicly available datasets.

Models

Model	Parameters	Notes
Qwen/Qwen3-VL-2B-Instruct	2B	Strong multimodal capabilities; main model used for fine-tuning
HuggingFaceTB/SmolVLM-256M-Instruct	256M	Smaller model for comparison; limited performance on this task

Qwen3-VL-2B was chosen for main experiments due to its larger capacity and better multimodal reasoning.

Datasets

Dataset	Description	Train	Validation	Test
linxy/LaTeX_OCR	Synthetic LaTeX + handwritten samples	1000	70	70
deepcopy/MathWriting-human	Real handwritten formulas	1000	70	70

For the mixed dataset setup, both LaTeX_OCR and MathWriting subsets were combined for supervised fine-tuning.

Training Setup

Hyperparameters and configuration:

- Fine-tuning method: LoRA

r = 8, α = 16, target modules = ["q_proj","k_proj","v_proj","o_proj"]

- Quantization: 8-bit (BitsAndBytes)
- Gradient checkpointing enabled
- Batch size: 2 (effective batch = 2 × 4 gradient accumulation = 8)
- Learning rate: 2e-4
- Warmup steps: 50

- Epochs: 12 (with early stopping)
- Max sequence length: 1024
- Evaluation: every 100 steps
- Device: GPU rtx 5070 laptop

Evaluation Metrics

To quantify the quality of LaTeX generation:

Metric	Description
BLEU	Measures n-gram overlap with reference LaTeX
ROUGE-L	Measures longest common subsequence (text-level similarity)
Character Accuracy	Fraction of correctly generated characters

Character-level accuracy is critical due to LaTeX's strict syntax.

Experimental Results

Model / Setup	BLEU	ROUGE-L	Char Accuracy
SmolVLM-256M Zero-shot	0.1070	0.2959	0.0310
SmolVLM-256M One-shot	0.0287	0.0876	0.0255
Qwen3-VL Zero-shot	0.6002	0.8467	0.0831
Qwen3-VL One-shot	0.2264	0.4605	0.0445
Qwen3-VL SFT LaTeX_OCR	0.8857	0.9594	0.6363
Qwen3-VL SFT LaTeX_OCR + MathWriting	0.8151	0.9530	0.5969

Training Curves

Training loss started high (~22) and rapidly decreased to plateau (~4.2)

Validation loss was lower than training (~3.2) due to cleaner data

Smoothed loss shows stable convergence

Overfitting indicator remains near zero → no overfitting observed

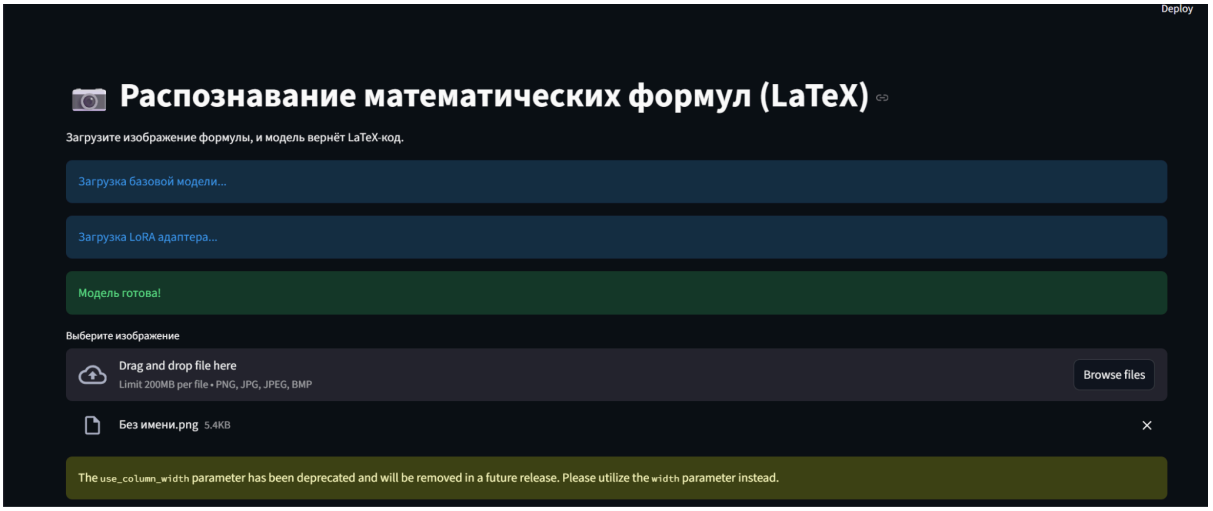
Observations

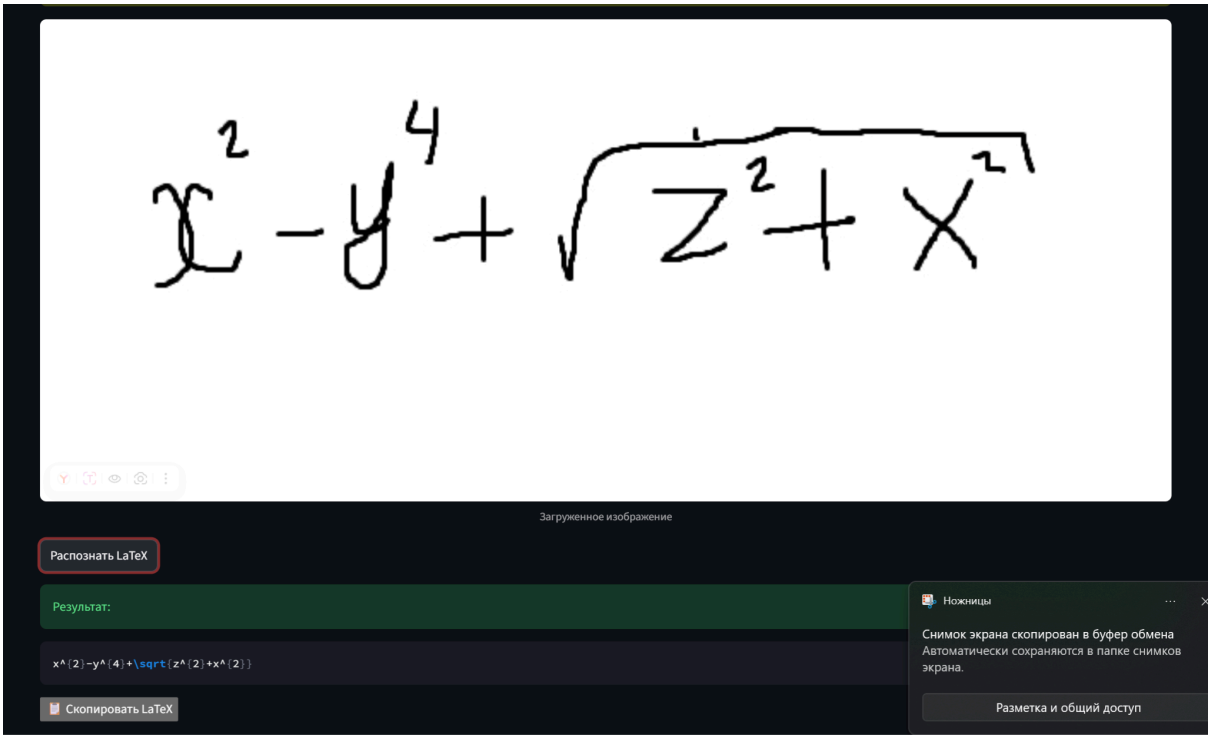
Domain shift: Mixed dataset (synthetic + handwritten) slightly reduces performance.

LoRA efficiency: 8-rank adapter allows fine-tuning large model on limited GPU memory.

One-shot prompt: performance worse than zero-shot due to distribution mismatch.

Task 2: Streamlit Application





The `use_column_width` parameter has been deprecated and will be removed in a future release. Please utilize the `width` parameter instead.

$$0{,}00\theta_6 = \frac{2{,}813}{R_6 + 33}$$

Загруженное изображение

Распознать LaTeX

Результат:

```
0,00\theta_{6}=\frac{2,813}{R_{6}+33}
```

Скопировать LaTeX

Выберите изображение

Drag and drop file here

Limit 200MB per file • PNG, JPG, JPEG, BMP

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The `use_column_width` parameter has been deprecated and will be removed in a future release. Please utilize the `width` parameter instead.

$$w\frac{d^2}{dt^2}y(t) + \beta \cdot \frac{d}{dt}y(t) + y(t) = 0$$

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w\frac{d^{2}y(t)}{dt^{2}}+\beta\cdot\frac{d}{dt}y(t)+y(t)=0
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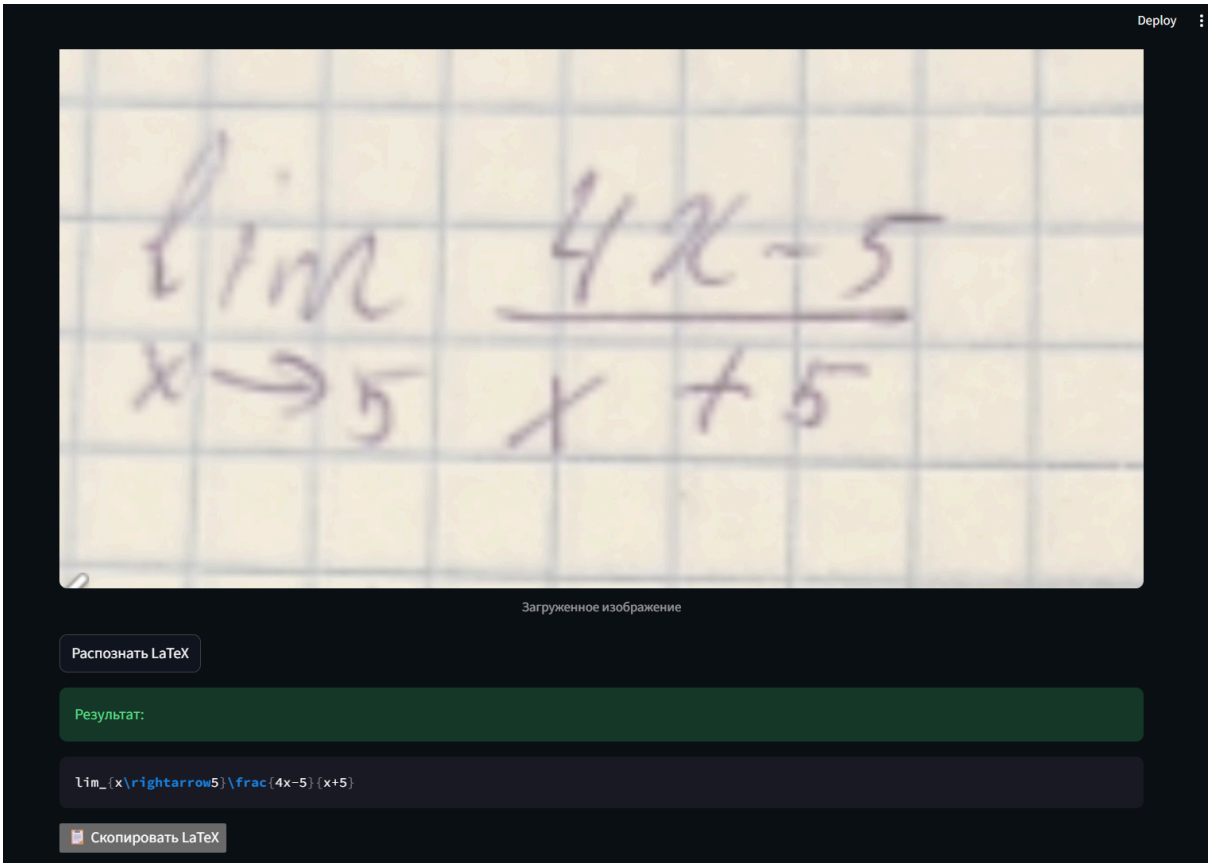
Скопировать LaTeX



Conclusion

Large VLMs such as Qwen3-VL-2B can be effectively fine-tuned to convert handwritten formulas into LaTeX.

Fine-tuning on a clean dataset (LaTeX_OCR) yields highest accuracy.

Mixed datasets introduce domain variability, slightly degrading performance.

LoRA and 8-bit quantization allow training on limited hardware.

Zero-shot and one-shot inference are feasible but much less accurate.

https://github.com/van-yak/Latex_project/tree/main

model and checkpoint: https://disk.yandex.ru/d/fdja_pUCrXxy9A